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ORIGINAL ARTICLE

10-week suspension and traditional push-up training: Comparison the effects on physical performance in young men

Entraînement de 10 semaines à la suspension et traditionelles aux pompes: Comparaison des effets sur les performances physiques chez les jeunes hommes

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KEYWORDS
Suspension training;
Traditional training;
Push-up;
Physical performance;
Healthy young men

Summary
Objectives. — Suspension training has become popular in sports and functional performance in recent years. Suspension training may promote higher improvement in physical performance compared to traditional training because of its greater instability. Therefore, this study aimed to compare the effects of 10-week suspension and traditional push-up training on physical performance in young men.
Methods. — Thirty-two young men were divided randomly into two groups: suspension push-up ($n = 17$) and traditional push-up ($n = 15$). Physical performance tests included 90^0 push-up test for strength-endurance, Closed Kinetic Chain Upper Extremity Stability test (CKCUES) for shoulder stability, Y Balance Test (YBT) for dynamic balance, and Seated Medicine Ball Throw Test (SMBT) for power. Assessments were completed in two sessions in both groups before and after the 10-week push-up training.
Results. — In baseline values, a significant difference was found for only the CKCUES test in favor of the suspension push-up group ($P < 0.05$). Both groups showed significant improvements in physical performance tests ($P < 0.05$); however, when compared groups, there was a significant difference in the 90^0 push-up test in favor of the suspension push-up group ($P < 0.05$). After 10 weeks of push-up training, Cohen's d effect size of both groups was high except for SMBT in the traditional push-up group.

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Conclusion. — Both suspension and traditional push-up training effectively improved physical performance in healthy untrained young men. However, suspension push-up is more efficacious than traditional push-up in strength-endurance. Our results suggest that healthy untrained young men who desire to increase muscular strength and endurance may prefer suspension push-up training.
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MOTS CLÉS
Entraînement en suspension ;
Entraînement en traditionnel ;
Pompes ;
Performance physique ;
Jeunes hommes en bonne santé

Résumé
Objectifs. — L'entraînement en suspension est devenu populaire ces dernières années dans le domaine du sport et de la performance fonctionnelle. L'entraînement en suspension peut favoriser une amélioration plus importante des performances physiques par rapport à l'entraînement traditionnel en raison de sa plus grande instabilité. Par conséquent, cette étude visait à comparer les effets d'un entraînement en suspension de 10 semaines et d'un entraînement traditionnel aux pompes sur les performances physiques de jeunes hommes..
Méthodes. — Trente-deux jeunes hommes ont été divisés au hasard en deux groupes : pompes en suspension (n = 17) et pompes traditionnelles (n = 15). Les tests de performance physique comprenaient le 90° push-up test pour la force-endurance, le CKCUES (Closed Kinetic Chain Upper Extremity Stability test) pour la stabilité des épaules, le Y Balance Test (YBT) pour l'équilibre dynamique et le SMBT (Seated Medicine Ball Throw Test) pour la puissance. Les évaluations ont été réalisées en deux séances dans les deux groupes, avant et après l'entraînement de 10 semaines aux pompes.
Résultats. — En ce qui concerne les valeurs de base, une différence significative a été constatée uniquement pour le test CKCUES en faveur du groupe des pompes en suspension ($P < 0,05$). Les deux groupes ont montré des améliorations significatives dans les tests de performance physique ($P < 0,05$) ; cependant, lorsque l'on compare les groupes, il y a une différence significative dans le test des 90° pompes en faveur du groupe des pompes en suspension ($P < 0,05$) . Après 10 semaines d'entraînement aux pompes, la taille de l'effet de Cohen des deux groupes était élevée, à l'exception du SMBT dans le groupe des pompes traditionnelles..
Conclusion. — L'entraînement aux pompes en suspension et l'entraînement traditionnel améliorent efficacement les performances physiques de jeunes hommes en bonne santé et non entraînés. Cependant, les pompes en suspension sont plus efficaces que les pompes traditionnelles en ce qui concerne la force musculaire et l'endurance. Nos résultats suggèrent que les jeunes hommes en bonne santé et non entraînés qui souhaitent augmenter leur force musculaire et leur endurance peuvent préférer l'entraînement aux pompes en suspension.Traduit avec www.DeepL.com/Translator (version gratuite).
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1. Introduction

Bodyweight training is strength training that allows an effective increase in muscle mass [1]. Suspension training, a new type of instability training, has become popular in athletics and fitness training in recent years [2—4]. Suspension training provides bodyweight exercises using two separately moving slings suspended by two straps with a fixed anchor position above the exerciser [5]. Suspension training is deemed a helpful method for enhancing neuromuscular activation that reviewed the use of heavy loads in traditional exercises [6]. Sports specificity increases with the addition of instability to bodyweight exercises and a high level of functional performance and gains are supported in suspension training [7]. The unstable nature of suspension training puts more strain on the neuromuscular system than traditional resistance exercises performed on stable surfaces and provides a strength increase, which may result from changes in muscle cross-sectional area and neuromuscular coordination [5,8—10]. Suspension training requires high motor control demands and the generation of muscle force necessary to overcome the load and maintain stability, which may contribute to the promotion of functional performance [11]. Suspension training has been added to traditional exercises to promote neuromuscular coordination and muscle recruitment, but there is debate concerning the effects of this combination on physical performance [6].

Push-up training has been studied previously for the exercises of the upper extremities and the trunk-stabilizing muscle performance [4,6,12,13]. Push-up is a traditional exercise that can be performed with a suspension device and used to train the shoulder musculature [12]. Challenge in maintaining the dynamic balance and stabilization in the suspension push-up, and also because of the exposure to the greater loads by the upper extremity, suspension push-up might be harder to perform [14]. Unstable surface training was previously carried out for rehabilitation and prevention programs; this type of training is currently included in strength and endurance programs due to the proprioceptive and neuromuscular gains they provide [6]. Unstable surfaces used during push-up training may also trigger motor learning adaptations to improve dynamic

balance and strength output [11], and this forms the basis of clinical information supporting the use of suspension systems [15]. It is possible that suspension training promotes greater improvement in functional performance compared to traditional training because of its higher instability [16]. Most studies comparing traditional strength training and suspension strengthening training have executed protocols on athletes and non-athletes because of the need for high-level of strength [17]. A previous systematic analysis reported that unstable surface training led to changes in muscle strength, power, and muscular speed parameters in young adults compared to traditional training [6]. Using unstable surfaces during strength training without any disadvantages compared to stable surfaces is strongly recommended for healthy young people [18] and untrained young people [19]. Since the results in the literature are heterogeneous, and there is no criterion for assessing it in different populations, it seems necessary to expand the unstable surface training with performance variables [6]. For this purpose, a previous systematic analysis suggested that to improve physical performance parameters such as strength and balance, future studies should focus on the suspension systems in non-athletes healthy individuals [17].

The controversy in literature concerning the use and outcomes provided by suspension training, the individuals, and exercises created a dispute on physical performance outcomes. It is needed to determine whether performing exercises with an unstable surface presents further impacts on muscle strength and power measures compared to a stable surface [6]. Therefore, this study aimed to to compare the effects of 10-week suspension and traditional push-up training on physical performance in healthy young men. We hypothesized that suspension push-up training would improve physical performance better than traditional push-up.

2. Materials and methods

2.1. Participants

This study was a single-center and two-group randomized trial conducted at the local university. Fifty-six men initially volunteered to take part in the research study. However, due to failure to meet the inclusion criteria ($n = 21$) and irregular participation ($n = 3$), thirty-two healthy untrained young men were included in this study. Before enrolment, all potential participants were questioned about their medical and health history to determine their eligibility. Participants were also questioned about their physical activity, and all were recreationally active; however, they were not a part of any resistance training until six months before push-up training. The inclusion criteria were: 18—25 years of age, being male, not having any injury shoulder region in the last six months, not having resistant physical exercise in the upper quarter currently, having a body mass index (BMI) below thirty and not using protein powder in the last six months. Exclusion criteria were: doing regular resistive physical exercise for the upper extremity in the last six months, having had a disability or injury in the last six months, having an orthopedic, neurological, cardiovascular or any health condition that prevents training, and having a disability or injury in the last six months. Computer randomization was used in the distribution of the groups, and the participants who met the inclusion criteria were divided into two groups: suspension push-up ($n = 17$) and traditional push-up ($n = 15$). The study was performed under the principles of the Declaration of Helsinki and approved by the Clinical Research and Ethics Committee of the author's affiliated institution. All participants were informed verbally and in writing about the study protocol and a written consent.

2.2. Study design

Participants' physical performances were assessed before and after 10-week push-up training. Assessments took place in two sessions in both groups. In the first session, shoulder stability and dynamic balance assessments were performed. Two days after, on the second session, strength-endurance and power assessments were performed. Participants were thoroughly encouraged to do ''as fast as possible'' for each pysical performance assessment. This study was single-blinded, in that the assessor performing the physical performance assessments were blinded to group allocation. While a blind evaluator made all the assessments to the groups, the other experimenter carried out the 10-week push-up training program.

2.3. Assessments

2.3.1. 90^0 push-up test

90^0 push-up test was used to evaluate the participants' upper extremity strength-endurance. For the test, participants took the push-up position at full extension of the elbows with the arms at shoulder width apart. The participants were asked to perform one push-up every three seconds. A metronome was used for the participants to follow the time. The test was terminated in the third second when the participant could not perform push-up, and the total number of push-ups was recorded (Fig. 1A). Test-retest reliability of the 90^0 push-up test has been previously reported [20].

2.3.2. Closed Kinetic Chain Upper Extremity Stability test

Closed Kinetic Chain Upper Extremity Stability test (CKCUES) is a simple and cheap test assessing shoulder stability. Two lines were drawn parallel to the participants' bodies for the test, about thirty-six inches apart from the surface. While the elbows were in full extension, the participant was asked to lift one hand and touch the other hand and then repeat the opposite (Fig. 1B). The number of repetitions performed within fifteen seconds was recorded. The test was repeated three times, and the mean value was recorded [21].

2.3.3. Y Balance Test

Y Balance Test (YBT) is a simple and reliable balance assessment method used to determine the dynamic balance [22]. To perform the YBT in the upper extremity, the participant is asked to reach with the free hand in the medial, inferolateral, and superolateral directions while maintaining weight bearing on the hand that is placed in a standardized location

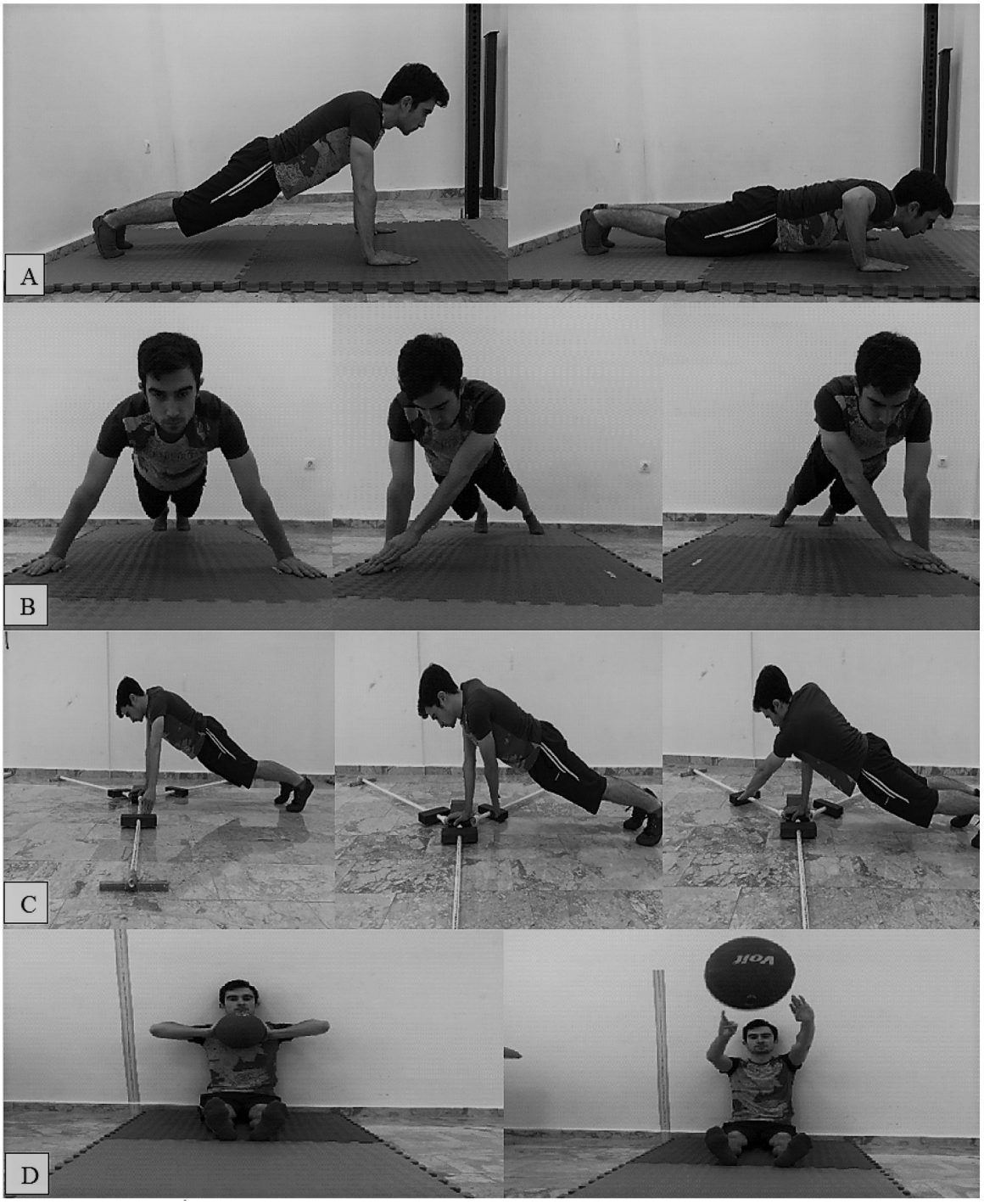


Figure 1 A. 90^0 push-up test. B. Closed Kinetic Chain Upper Extremity Stability test. C. Y Balance Test. D. Seated Medicine Ball Throw Test.

(Fig. 1C). Maximum reach distance was recorded as the average of three measurements. Higher reach scores indicate a better dynamic balance.

2.3.4. Seated Medicine Ball Throw Test

The Seated Medicine Ball Throw Test (SMBT) was used to assess the anaerobic power of the upper extremity. For the test, the participant was placed in a position with the legs in full extension and the back, shoulders, and head against the wall. The participant lifted the three-kilogram medicine ball with 90^0 flexion and abduction with both upper extremities and threw the medicine ball forward as far as long (Fig. 1D). Four throws were performed, and the average distance score was recorded [23].

2.3.5. Borg Category Ratio

The rating of perceived exertion was presented by Borg (Borg CR-10) as a novel approach to define adaptations to

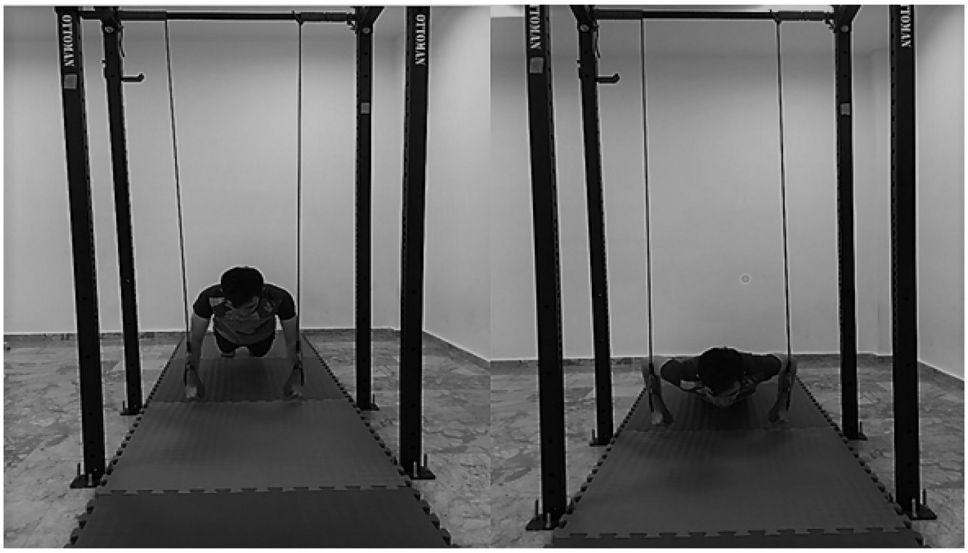


Figure 2 Suspension push-up training.

physical exercise in fitness and sports settings. Our study used a modified version of Borg's CR-10 scale to quantify the perception of push-up training exertion. The CR-10 scale is scored between 0 (rest), 1 (very, very easy), 2 (easy), 3 (moderate), 4 (somewhat hard), 5 (hard), 7 (very hard) and 10 (maximal) [24,25]. After 30 minutes of each training session, we questioned participants how much they exerted due to push-up training, and we asked participants to mark a number on the CR-10 scale to determine their overall exertion.

2.4. Experimental protocol

2.4.1. Suspension push-up

Redcord suspension system (Kilsund, Norway) was used for suspension push-up training. The Redcord device is an easy-to-carry, convenient, and effective strength training device. Redcord slings are suspended by two straps with a fixed anchor position above the exerciser. The slings were hanging at the shoulder opening while the push-up was being made, and the participant's hands were ten cm above the surface. A supervisor taught each participant the proper push-up form and counted the number of successful repetitions. The body form was monitored, and repetitions were repeated if the correct depth (90^0) was not acquired [19] (Fig. 2).

2.4.2. Traditional push-up

Traditional push-up was performed on a stable surface with shoulder-width apart, and fingers pointed forward. Participants were instructed to keep a straight spine and feet together during the movement. The body form was monitored, and repetitions were repeated if the correct depth (90^0) was not acquired [13].

2.4.3. Training program

The suspension and traditional push-up groups trained twice per week (48—72 hours apart) over 10weeks for a total of twenty sessions. A standard training program was applied to suspension and traditional push-up groups (12 repetitions $\times$ 3 sets) to monitor how the same training load affects perceived exertion [26]. The training program consisted of a warm-up, push-up training, and cooling periods (Table 1). A one-minute recovery period was maintained throughout the training conditions. For adaptation to the training program, repetitions and push-up training sets were kept low in the first two weeks and then increased in the third week. Since only one participant in the suspension push-up group could not do the regular push-up, the participant performed the push-up training in the modified push-up position. After the first two sessions of the adaptation period, the participant could able to push-up training in the regular position. Training sessions were conducted in a research laboratory under the direct supervision of the investigator, at the same time of day ($\pm$1 hour) for each participant and under controlled environmental conditions. We used a metronome to control the cadence of the push-up repetitions.

2.5. Statistical analysis

A two-way, repeated-measure analysis of variance was used to compare physical performance differences, with time serving as the within-group and training protocol as the between-group factor. Paired t-tests were used to identify significant differences before and after the in-group training. The Mann-Whitney U and Wilcoxon tests were used if data did have not normal distributions. A Cohen's d effect size was also performed to demonstrate the magnitude of the training effect in each group. SPSS statistical software (v.22; SPSS, Chicago, IL) was used for all analyses, and the statistical significance was set at $P \leq 0.05$.

3. Results

The suspension push-up group consisted of seventeen participants (mean age, 22.05 ± 1.78 years; weight, 71.05 ± 12.28 kg; height, 177.29 ± 5.64 cm) and traditional

Table 1 Exercises in suspension and traditional push-up training program.

Warming period	
Jogging	5 minutes
Scapular retraction with shoulder elevation	10 repetitions
Shoulder flexion-extension	10 repetitions
Circles forward and back in 90^0 abduction of arms	10 repetitions
Shoulder horizontal abduction	10 repetitions
Stretching cervical lateral flexors	5 repetitions
10-week push-up training: suspension and traditional push-up	
1st week	1. Session = 8 repetitions × 2 sets
	2. Session = 10 repetitions × 2 sets
2nd week	3. Session = 8 repetitions × 3 sets
	4. Session = 10 repetitions × 3 sets
3rd—10th weeks	5—20th sessions = 12 repetitions × 3 sets
Cooling period	
Pectoralis major stretching	30 seconds × 2 repetitions
Shoulder internal and external rotator stretching	30 seconds × 2 repetitions
Stretching shoulder posterior capsule	30 seconds × 2 repetitions

push-up group consisted of fifteen participants (mean age, 20.80 ± 2.04 years; weight, 74.93 ± 9.42 kg; height, 178.60 ± 6.64 cm) and there was no difference in age, weight and height between groups ($P > 0.05$). In the first two sessions of the adaptation period, since only one participant in the suspension push-up group could not do the regular push-up, the participant performed the push-up training in the modified push-up position. As for physical performance data before the training program, only a significant difference was observed in favor of suspension push-up in the CKCUES compared with the traditional push-up ($P = 0.024$). No significant difference was found between groups for baseline values of 90^0 push-up test, YBT, and SMBT . After 10 weeks of push-up training, Cohen's d effect size of both groups was high except for SMBT in the traditional push-up group.

3.1. Strength and endurance

After training, a significant increase was showed in 90^0 push-up tests in both groups ($P < 0.001$, Table 2). Cohen's d effect size of groups was high (suspension push-up group: 2.10, traditional push-up group: 1.22, Table 2). When the difference values before and after training (Delta, △) of the 90^0 push-up test were compared, a significant difference was found in favor of the suspension push-up group ($P = 0.033$, Table 3).

3.2. Shoulder stability

After training, a significant increase occurred in CKCUES in both groups ($P < 0.001$, Table 2). Cohen's d effect size of groups was high (suspension push-up group: 1.68, traditional push-up group: 1.98 Table 2). When Delta values were compared, no significant difference was found between the groups ($P = 0.576$, Table 3).

3.3. Dynamic balance

After training, both groups revealed a significant increase in YBT ($P < 0.001$, Table 2). Cohen's d effect size of groups was high (suspension push-up group: 1.57, traditional push-up group: 2.47 Table 2). When Delta values were compared, no significant difference was found between groups ($P = 0.079$, Table 3).

3.4. Power

After training, a significant increase was observed in SMBT in both groups ($P < 0.05$, Table 2). Cohen's d effect size was high in the suspension push-up group; however the traditional push-up group were moderate (suspension push-up group: 1.04, traditional push-up group: 0.58, Table 2). When Delta values were compared, no significant difference was found between groups ($P = 0.523$, Table 3).

3.5. Perceived exertion

The mean perceived exertion levels in the adaptation period of the suspension push-up and traditional push-up group were 5.74 ± 2.54 and 4.58 ± 2.07, respectively ($P = 0.172$). After the adaptation period, the suspension and traditional push-up groups' perceived exertion levels increased from the 3rd week (suspension push-up group, 6.17 ± 1.46; traditional push-up group, 5.80 ± 2.48) ($P = 0.600$) and then gradually decreased until the end of the 10 weeks. There was a difference between groups only in the twentieth session on the CR-10 scale (suspension push-up group, 2.14 ± 2.46; traditional push-up group, 3.46 ± 1.94) ($P = 0.030$) (Table 4).

4. Discussion

The results of the current study, which aimed to compare the effects of 10 weeks of suspension and traditional push-up training on physical performance in healthy young men, showed that both suspension and traditional push-up

Table 2 Comparison the physical performance before and after training in suspension and traditional push-up groups.

Variables	SP ($n = 17$)				TP ($n = 15$)			
	Pre	Post	*P*	Cohen d	Pre	Post	*P*	Cohen d
90^0 PU test	19.29 ± 10.14	31.47 ± 12.19	0.0001* (t = −8.736)	2.10	19.40 ± 10.33	26.86 ± 12.60	0.0001* (t = −4.728)	1.22
CKCUES	24.41 ± 5.22	29.05 ± 3.49	0,0001* (z = −3.670)	1.68	21.73 ± 4.13	26.40 ± 3.86	0.0001* (t = −7.690)	1.98
YBT	91.57 ± 10.16	98.18 ± 8.05	0.0001* (t = −6.475)	1.57	94.00 ± 10.21	103.17 ± 9.75	0.0001* (t = −9.569)	2.47
SMBT	427.17 ± 51.07	446.50 ± 55.67	0.001* (t = −4.307)	1.04	424.20 ± 45.14	438.60 ± 52.77	0.039* (t = −2.273)	0.58

P: significance value; Cohen d: effect size; t: paired *t*-test; z: Wilcoxon test.
* Statistical significance.

Table 3 Comparison the Delta values of suspension and traditional push-up groups.

Variables	SP ($n = 17$)	TP ($n = 15$)	*P*	Cohen d
90^0 PU test	−12.18 ± 5.75	−7.47 ± 6.12	0.033* (t = 2.245)	0.79
CKCUES	−4.65 ± 2.76	−4.67 ± 2.35	0.545 (z = −0.605)	0.00
YBT	−6.60 ± 4.20	−9.16 ± 3.70	0.079 (t = −1.816)	0.64
SMBT	−19.32 ± 18.5	−14.4 ± 24.53	0.523 (t = −0.646)	0.22

P: significance value; Cohen d: effect size; t: independent samples t-test; z: Mann-Whitney U test.
* Statistical significance.

training were effectively improved physical performance in healthy young men. Suspension push-up training was further efficacious in strength-endurance than traditional push-up. However, suspension and traditional push-up training similarly affect on shoulder stability, dynamic balance, and power. Besides, the results showed that until the eighth week in suspension push-up and the 10th week in a traditional push-up, the level of exertion was somewhat hard or higher in both groups. The level of perceived exertion in suspension and traditional push-up training initially created a high level of difficulty; however, in the following weeks of the push-up training, the perceived exertion decreased in both groups in line with the loading principles.

Push-up training used to increase muscle strength and endurance using bodyweight is an effective training method of strengthening. In a previous study, the resistance training with bench press and suspension push-up for two to three sessions per week and a total of 13 weeks were compared, and the authors found that suspension push-up training was more effective in developing strength and endurance [19]. Similarly, two times per week, eight weeks of suspension push-up training resulted in better outcomes than traditional push-up training in endurance and fatigue [27]. On the other hand, the results of previous studies showed that suspension and traditional push-up training produced similar results in strength [28—30], power [28], and dynamic balance [29,31]. In our study, we found that suspension push-up training was further effective than traditional push-up in strength and endurance progress, which we had hypothesized. It is likely that during the suspension push-up training, three-dimensional stability in the shoulder may force the muscles further than traditional push-up [4,12,32], which might be the reason for this result.

Shoulder stability is ensured by the compatibility of static and dynamic factors and their sufficient strength and balance. Suspension training has been reported to increase the the biceps brachii muscle's effectiveness in the role of shoulder stabilization and motor control [27]. In addition, suspension training requires high motor control and necessary muscle power generation to maintain stability, and this contributes to improving functional performance [16]. Resistance training on an unstable surface can force the extremity musculature to play a great function in joint stability [11]. In our study, although the suspension push-up group had higher initial shoulder stability values after random allocation, the differences observed in shoulder stability outcomes after push-up training were similar, indicating that the result is not in consonance with our hypothesis. This may indicate that participants in the suspension push-up group, who had better shoulder stability at baseline, might be more likely to experience significant changes in shoulder stability with 10 weeks of training.

Dynamic balance is of substantial importance for activities that require high performance. Exercise on unstable surfaces is thought to strain the synergistic and stabilizing muscles [33], and the suspension system forces each arm

Table 4 Comparison of the perceived exertion level (CR-10) in suspension and traditional push-up groups during 10-week (20 sessions) push-up training.

CR-10	Suspension push-up ($n = 17$)	Traditional push-up ($n = 15$)	P
	Mean ± SD	Mean ± SD	
Session 1	6.35 ± 2.93	4.53 ± 2.47	0.070 (t = −1.881)
Session 2	5.88 ± 2.20	4.33 ± 2.12	0.053 (t = −2.016)
Session 3	5.44 ± 2.88	4.86 ± 2.16	0.533 (t = −0.630)
Session 4	5.29 ± 2.44	4.60 ± 2.26	0.413 (t = −0.830)
Session 5	6.17 ± 1.46	5.80 ± 2.48	0.600 (t = −0.529)
Session 6	5.47 ± 2.06	5.53 ± 1.95	0.863 (z = −0.172)
Session 7	5.05 ± 1.91	5.06 ± 1.66	0.908 (z = −0.115)
Session 8	4.35 ± 2.28	5.33 ± 2.05	0.108 (z = −1.607)
Session 9	4.52 ± 1.73	4.93 ± 1.83	0.527 (t = 0.640)
Session 10	4.76 ± 2.01	4.86 ± 2.23	0.893 (t = 0.136)
Session 11	4.17 ± 2.12	4.53 ± 2.06	0.428 (z = −0.793)
Session 12	4.23 ± 2.10	4.26 ± 1.98	0.939 (z = −0.077)
Session 13	4.11 ± 2.17	4.13 ± 1.68	0.758 (z = −0.308)
Session 14	4.05 ± 1.88	4.20 ± 1.52	0.604 (z = −0.519)
Session 15	3.52 ± 1.46	4.00 ± 1.41	0.261 (z = −1.124)
Session 16	3.41 ± 2.01	4.06 ± 1.38	0.078 (z = −1.674)
Session 17	3.29 ± 2.11	4.13 ± 2.09	0.097 (z = −1.658)
Session 18	3.23 ± 1.67	4.13 ± 1.40	0.062* (z = −1.867)
Session 19	2.94 ± 1.59	3.93 ± 1.79	0.057 (z = −1.905)
Session 20	2.14 ± 2.46	3.46 ± 1.94	0.030* (z = −2.175)

z: Mann Whitney U test, t: independent samples t-test.
* $P < 0.05$ statistically significant difference.

of the individual to use independently of each other in order to keep a dynamic balance [27]. When using an unstable surface during exercise in the upper extremity, synergistic activation of not only the shoulder muscles but also the abdominal muscles may be necessary to keep the whole body in a stable and required position [34]. Previous studies stated that 8-week [29] and 7-week [30] unstable surface and traditional strengthening training improve balance but do not have superiority over each other. In addition, it has been shown that 13 weeks of suspension training and traditional resistance exercises provide similar results in dynamic balance development [19]. Considering the contribution of suspension and traditional push-up training to dynamic balance, the result obtained in our study is consistent with the findings in the literature and shows that both suspension and traditional push-up training increase dynamic balance but are not superior to each other. However, this result is not in line with our hypothesis.

Although exercise on unstable surfaces can cause a drop in power output, it is generally considered a

fundamental part of a versatile exercise program [34]. For non-elite athletes, current literature confirms that unstable surface training creates adequate strength and power to ensure their training adaptation [6]. It is stated that gaining strength with a light to moderate load may be more effective for power development than using a heavy load [35]. However, exercises performed on unstable surfaces are less effective in strength development than those performed on a stable surface [6]. Our results indicated that both suspension and traditional push-up training had similar improvements in power, which is not in line with our hypothesis. Obtaining similar power adaptations in suspension and traditional push-up groups due to the use of body weight in 10-week push-up training, which is one of the basic theories of suspension training, may explain the result of our study.

Suspension training can be classified as moderate-to-vigorous intensity training, which can stimulate physical performance [5]. Perceived exertion alters with the number of muscle groups and joints involved in the movement and the intensity of load [6,36,37]. Training at a high intensity with fewer repetitions is perceived as more challenging than exercising at low intensity with higher repetitions [36]. Besides, strength training with low loads compared to heavy loads is deemed equally efficient at enhancing muscle strength and power in healthy young adults [6]. Based on this background, we aimed to adapt sedentary young participants to exercise by applying low repetition and low-intensity training at the beginning of this study. We applied a standard program to both suspension and traditional push-up groups (12 repetitions $\times$ 3 sets) to monitor how the same training load affects perceived exertion. Our results showed that the level of perceived exertion in suspension and traditional push-up training initially created a high level of difficulty on the CR-10 scale. However, both groups' perceived exertion levels decreased in the following weeks of the push-up. Until the eighth week in suspension push-up and the tenth week in a traditional push-up, the level of exertion was 4 (somewhat hard) or higher in both groups. This level of exertion indicates that the training load is forcing the participants in line with the loading principles. Moreover, the participant's perception of fatigue may influence perceived exertion during push-up training. Despite the possible effects, the literature proposes that the perceived exertion as an indicator of exercise intensity is valid [38].

A randomized, single-blinded, limited age range and 10-week long-term duration of push-up training adaptations are the strengths of our study. Since we did not use any resistance equipment, we could raise the number of sets and repetitions, which is considered a limitation of our study. Most studies on perceived exertion have investigated the relationship between one repetition of maximum or maximal voluntary contraction. As we did not use these measurements, that could be another limitation of our study.

5. Conclusion

The current study showed that suspension and traditional push-up training were effectively improved physical performance in healthy untrained young men. However, our hypothesis was partially confirmed with strength and endurance performance being better in the suspension push-up group, but no difference was seen in dynamic balance, shoulder stability, and power. The perceived exertion in suspension and traditional push-up training initially created a high level of difficulty in both groups; however, in the following weeks of the training, the level of exertion was decreased to 4 (somewhat hard) or higher level in both groups, which indicates that the training load is forcing the participants in line with the loading principles. We did not find a study assessing bodyweight training and shoulder stability in suspension system when we reviewed the literature. Shoulder stability was the first time compared in suspension and traditional push-up training in healthy untrained young men. Increased shoulder stability of young men in both groups supports the idea that suspension and traditional push-up training could be used to increase shoulder stabilization. Our results might be a lead for future studies in this field. In conclusion, this study provides information for the practitioners who train to improve performance in healthy untrained young men. Our results suggest that healthy untrained young men who desire to increase muscular strength and endurance may prefer suspension push-up training.

Disclosure of interest

The authors declare that they have no competing interest.

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